Advanced AC Circuit

Laboratory Exercise No: 3

Measurement for Transfer Function

Introduction

Driving point impedance is defined as the ratio of phasor voltage to its current at the same side of a two-port network. Transfer function such as voltage gain characterizes the ratio of output (voltage) and input (voltage) in phasor domain. Such a function varies with frequency, which is the basis of control theory in communication.

The main objective of this laboratory exercise is to test the characteristics of the frequency response of AC circuits. In this lab, you will learn how to use oscilloscope to measure voltage, current and phase difference at different frequencies and to plot the amplitude response and phase response.

1. Driving point function and transfer function

• Build the circuit shown in Fig.1 with **Function Waveform Generator** as voltage source v_S. Use **Digital Oscilloscope** to observe the input voltages and output currents from the source, and measure the maximum value (or rms) of voltages (currents are actually measured by voltages across a resistor) and phase differences at the selected frequencies. Driving point impedance then can be calculated at each frequency.

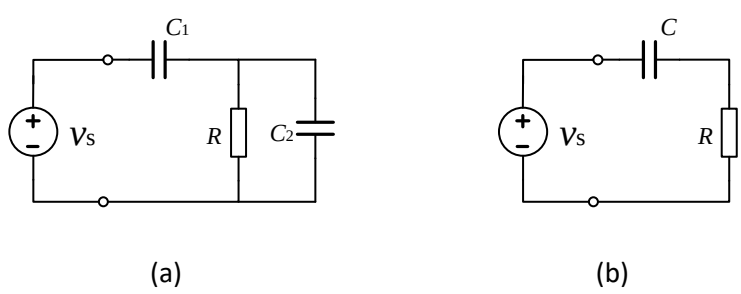

Figure 1: Circuits in parallel (a) and in series (b)

• Build the circuit shown in Fig.2 with **Function Waveform Generator** as voltage input v_S and voltage v_O as output. Use **Digital Oscilloscope** to observe and measure the value of voltages and phase differences at the selected frequencies. Then transfer voltage gain can be calculated at each frequency.

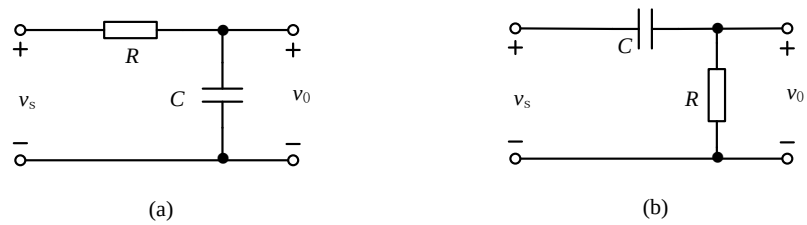

Figure 2: Circuits in low-pass (a) and in high-pass (b)

2. Pre-Lab Work

a) **Theory analysis**

• Use phasor analysis method to determine the driving point impedance shown in Fig.1(a) and (b), respectively. Given R=1.2kΩ, C1=0.47μF, C2=0.047μF in Fig.1(a), calculate the amplitude and the phase of the impedance at frequency 500Hz, 2kHz, 4kHz and 10kHz. Given R=510Ω, C=0.1μF in Fig.1(b), repeat the calculation above at frequency 200Hz, 2kHz, 5kHz and 10kHz.

• Use phasor analysis method to determine the voltage gain in the circuit shown in Fig.2(a) and (b), respectively. Given R=5.1kΩ, C=0.047μF, calculate the amplitude of the transfer function and its phase with respect of frequency. Express the amplitude in dB.

b) **Simulation Practice**

Use Multisim 14.0 to simulate your circuits. Employ a **Bode Plotter** to observe frequency response. Connect the input signal to 'in' connector and output signal to 'out' connector. Set 'Mode' to 'Magnitude' or 'Phase'. Set measure type on 'Horizontal' and 'Vertical' to log or linear. Set the range of frequency. Run the software and observe the plot. Use curser to find the corresponding amplitude and phase of the transfer function in Fig.1 and Fig.2 at given frequencies. Don't forget to find out the frequency at which the voltage gain is -3dB (or 0.707) in Fig.2. Compare the results with those that you have calculated. Output the Bode plots of each transfer function as a record.

Method of measuring current

In practical circuit, current is often measured indirectly because oscilloscope can only accept voltage signal. A resistor with small resistance can be introduced into circuit where the current is to be measured. Note that one of the terminals of the resistor should be connected to the reference earth so that the other end is the node whose voltage can be measured by oscilloscope. Then the current can be achieved by Ohm's Law. Since the introduction of the resistor also introduces error to the voltage measuring, the resistance should be selected as small as possible (far less than the impedance of the circuit) but too small resistance may interfere the result of the measurement.

3. Lab Section 1

In this laboratory, you will learn how to use oscilloscope to measure driving point impedance. Build the circuit shown in Fig.1 on breadboard. Select a resistor with small resistance (if needed) to measure its voltage and then apply phasor Ohm's Law to get its current. The procedure includes the following steps:

(i) Measure the resistance and capacitance rated R=1.2 kΩ, C=0.47μF, C2=0.047μF.

(ii) Build the circuit shown in Fig.1(a) and then (b) on breadboard and connect the instruments correctly.

(iii) Select Sine Function from **Function Waveform Generator** as v_s, and monitor it on the **Digital Oscilloscope** (CH1). Use CH2 to monitor current which goes through v_s.

(iv) Observe the frequency response roughly to see if you have done it correctly. Set a low frequency on the source and increase it. Adjust **Digital Oscilloscope** to show stable waveforms at lower frequency and then higher frequency to see if it performs the property of low-pass or high-pass.

(v) Measure the impedances at the setting frequencies and compare them with theoretical values and simulation results.

Requirements and notes

(i) While measuring the impedances, every time you change the frequency, make sure to adjust v_s so that its output voltage keeps the same.

(ii) Use double traces method and ellipse method to measure phase difference at each frequency point.

(iii) Fill all the results in the properly designed form.

(iv) Record the waveform and ellipse shape from Fig.1(b) at 2kHz on which the original data should be shown.

4. Lab Section 2

In this laboratory, you will learn how to use oscilloscope to measure transfer function amplitude in dB and Np. Build the circuit shown in Fig.2 on breadboard. The procedure includes the following steps:

(a) Measure the 1st order RC low-pass frequency characteristics:

(i) Build the circuit shown in Fig.2(a) on breadboard and connect the instruments correctly.

(ii) Select Sine Function from **Function Waveform Generator** as input, set the voltage to 1V and monitor it on the oscilloscope (CH1). Feed the output voltage into CH2.

(iii) Observe the frequency response roughly to see if you have done it correctly. Set the frequency from 50Hz to 20kHz. Adjust **Digital Oscilloscope** to show stable waveforms at lower frequency and then higher frequency to see if it performs the property of low-pass.

(v) Measure and record the cutoff frequency at -3dB and then measure other frequencies one by one. Frequency response is supposed to be measured in 'dB' and phase response in 'degree'. Record all the data in a form you designed by yourself.

(b) Measure the 1st order RC high-pass frequency characteristics:

(i) Build the circuit shown in Fig.2(b) on breadboard and connect the instruments correctly.

(ii) Select Sine Function from **Function Waveform Generator** as input, set the voltage to 1V and monitor it on the oscilloscope (CH1). Feed the output voltage into CH2.

(iii) Observe the frequency response roughly to see if you have done it correctly. Set the frequency from 20kHz to 50Hz. Adjust **Digital Oscilloscope** to show stable waveforms at higher frequency and then lower frequency to see if it performs the property of high-pass.

(iv) Measure and record the cutoff frequency at -3dB and then measure other frequencies one by one. Frequency response is supposed to be measured in 'Np' and phase response in 'degree'. Record all the data in a form you designed by yourself.

Requirements and notes

(i) While measuring the transfer function, every time you change the frequency, make sure to adjust v_s so that its output voltage measured on the oscilloscope keeps the same, 1V.

(ii) Select the measuring frequency reasonably according to how characteristic curve changes. The number of frequency points should not be less than 10. You'd better plot the points on paper to make a curve as you are doing the measurement. Once you find out that the points are not enough to plot the characteristics, measure more in time.

5. Submission

a) Complete the theory analysis including equation deductions, measuring circuits and calculations;

b) Complete the software simulation including the simulating circuits, magnitude and phase response plots, measurement results at each frequency, and output waveforms;

c) Complete the practical measurements including data filling in forms, waveforms output and frequency response plots (in semi-logarithm);

d) Answer the following questions:

1) While measuring the driving point impedance in Fig.1(b), is a small resistor needed to measure the current? Why?

2) In the measurement, how to determine the -3dB frequency?

3) While measuring the driving point impedances, use the Digital Multi-meter to replace the Digital Oscilloscope in Fig.1(a) and (b). Are there any differences in the use of two instruments?

e) Make sure the names and IDs of **BOTH** team members are on the report,

g) Your prelab report should be submitted to Moodle before your lab hour. You should submit the completed report to Moodle by 12:00pm a week later after your lab day.

6. Assessment

i) This lab is assessed by 100 marks.

ii) The prelab work is assessed individually by 30% which includes theory analysis and simulation work.

iii) The lab work is designed to be completed in groups of two. Each group is required to submit ONE report which is assessed by 70%. Personal prelab work should be attached to the report behind. And this part of work should still be in the report but in a better way.